

# Earth-observation missions, data levels and SNAP operators

Reference tables for the Arischia methane-budget project · V. Emberumal 2248220 · Earth Observation, Prof. F. Nunziata

Rows with a **thick left rule and grey fill** are the ones actually used in this project. Everything else is context the examiner may ask about.

## 1 THE FIVE MEASUREMENT FAMILIES — EVERY SENSOR HERE IS ONE OF THESE

| FAMILY                    | SOURCE OF THE SIGNAL                 | WHAT IT FUNDAMENTALLY MEASURES                                                                   | EXAMPLE HERE     |
|---------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------|------------------|
| Passive optical           | reflected sunlight                   | Surface reflectance in fixed bands. Needs daylight, blocked by cloud and smoke.                  | Sentinel-2 MSI   |
| Thermal infrared          | the surface's own emission           | Radiance at 3–12 μm → brightness temperature via Planck. Works at night.                         | MODIS B21/B31    |
| Spectrometer (absorption) | reflected sunlight, finely dispersed | Depth of molecular absorption lines → column amount of a gas.                                    | TROPOMI SWIR     |
| Active microwave (SAR)    | its own transmitted pulse            | Backscattered power and <b>phase</b> . Sensitive to roughness, moisture, structure. All-weather. | Sentinel-1 C-SAR |
| Lidar / altimeter         | its own laser or radar pulse         | Two-way travel time → range, hence height or surface elevation.                                  | GEDI, Poseidon-4 |

## 2 THE COPERNICUS SENTINEL FAMILY

| MISSION                        | INSTRUMENT             | MEASUREMENT PRINCIPLE                                                                             | BANDS / FREQUENCY                 | RESOLUTIO<br>N                           | REVISIT           | PRIMARY ROLE                                                                                          | HERE?         |
|--------------------------------|------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------|---------------|
| Sentinel-1<br>A / B / C / D    | C-SAR                  | Active microwave. Transmits its own pulse and times the echo — day, night and through cloud.      | C-band 5.405 GHz<br>VV+VH / HH+HV | IW: 5 × 20 m<br>GRD<br>2.3 × 14 m<br>SLC | 6–12 d            | Deformation, flood, ice, soil moisture, structure                                                     | tried, failed |
| Sentinel-2<br>A / B / C        | MSI                    | Passive optical push-broom. Measures reflected sunlight in 13 fixed spectral bands.               | 443–2190 nm<br>13 bands           | 10 / 20 / 60 m                           | 5 d<br>(two sats) | Land cover, vegetation, <b>burn severity</b> , agriculture                                            | <b>core</b>   |
| Sentinel-3<br>A / B            | OLCI, SLSTR, SRAL, MWR | Optical + thermal radiometry and radar altimetry. SLSTR views each scene twice (nadir + oblique). | OLCI 21 bands<br>SLSTR 9 + 2 fire | OLCI 300 m<br>SLSTR 0.5–1 km             | < 2 d             | Ocean/land colour, surface temperature, <b>FRP product</b>                                            | no            |
| Sentinel-4                     | UVN on MTG-S           | Geostationary UV–VIS–NIR sounder — stares at Europe rather than orbiting past it.                 | 305–775 nm                        | ≈ 8 km                                   | hourly            | Air quality over Europe, diurnal cycle of pollutants                                                  | no            |
| Sentinel-5P                    | TROPOMI                | Passive push-broom grating spectrometer. Molecular absorption in reflected sunlight.              | 270–2385 nm<br>4 spectrometers    | 5.5 × 7 km<br>(CH <sub>4</sub> )         | daily             | Trace gases: CH <sub>4</sub> , CO, NO <sub>2</sub> , O <sub>3</sub> , SO <sub>2</sub> , HCHO, aerosol | <b>test</b>   |
| Sentinel-5                     | UVNS on MetOp-SG A     | Operational follow-on to S-5P on the polar meteorological platform.                               | 270–2385 nm                       | 7 × 7 km                                 | daily             | Long-term operational atmospheric composition                                                         | no            |
| Sentinel-6<br>Michael Freilich | Poseidon-4, AMR-C      | Radar altimetry: two-way pulse travel time to the sea surface, corrected for wet troposphere.     | Ku + C band                       | along-track ≈ 300 m                      | 10 d              | Global mean sea level, ocean topography                                                               | no            |

## 3 SENTINEL EXPANSION MISSIONS — WHAT IS COMING, AND WHY IT MATTERS TO THIS PROJECT

| MISSION        | WHAT IT ADDS                                                                                             | RELEVANCE TO A FIRE METHANE BUDGET                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO2M           | Imaging spectrometer for CO <sub>2</sub> and CH <sub>4</sub> at ≈ 2 × 2 km with a much lower noise floor | The first Copernicus mission that could plausibly close the gap this project measured — a 0.88 ppb signal against an 11 ppb noise floor.                |
| CHIME          | Hyperspectral VNIR–SWIR, ≈ 30 m, hundreds of contiguous bands                                            | Species-level fuel discrimination: would replace the CORINE four-class fuel type with a measured composition, attacking the ±40 % emission-factor term. |
| LSTM           | Land surface temperature at 50 m, high revisit                                                           | Thermal at a resolution matched to the fire, instead of MODIS at 1 km — would remove the three-pixel limitation of the FRP route.                       |
| ROSE-L         | L-band SAR, longer wavelength than Sentinel-1 C-band                                                     | Penetrates canopy, so structural change and biomass loss become observable where C-band saturates.                                                      |
| CRISTAL / CIMR | Polar ice altimetry, passive microwave imaging radiometry                                                | Cryosphere; no direct relevance here.                                                                                                                   |

**One-line answer if asked "why Sentinel-2 and not Landsat?"** — Landsat 8/9 OLI has the same NIR and SWIR bands and a longer archive, but 30 m instead of 10–20 m and a 16-day revisit per satellite. Over a 14-day fire that is at best one clear pair; Sentinel-2 gave me several.

#### 4 OTHER SATELLITES — USED, CONSIDERED, AND REJECTED

| MISSION / SENSOR     | TYPE                                           | RESOLUTION & REVISIT                              | WHAT IT MEASURES                                                                               | VERDICT IN THIS PROJECT                                                                                                                                                                                                                                                                                                         |
|----------------------|------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MODIS Terra + Aqua   | Passive optical + thermal, whisk-broom scanner | 250 m / 500 m / 1 km<br>4 overpasses per day      | 36 bands. B21/B22 at 3.96 $\mu\text{m}$ and B31 at 11 $\mu\text{m}$ give fire radiative power. | <b>Used.</b> FRP $\rightarrow$ FRE $\rightarrow$ biomass burned. The thermal audit that is physically independent of the optical route.                                                                                                                                                                                         |
| VIIRS S-NPP, NOAA-20 | Passive optical + thermal scanner              | 375 m fire channels twice daily                   | Same physics as MODIS with a 375 m I-band fire product.                                        | Not used. Would give $\approx 7\times$ more fire pixels — the single cheapest upgrade to the FRP route.                                                                                                                                                                                                                         |
| Landsat 8 / 9        | Passive optical (OLI) + thermal (TIRS)         | 30 m optical, 100 m thermal<br>16 d per satellite | NIR and SWIR equivalents of B8/B12; TIRS gives surface temperature.                            | Not used. Same indices, coarser and slower; valuable only for pre-2015 baselines.                                                                                                                                                                                                                                               |
| GEDI (on the ISS)    | Full-waveform lidar, 1064 nm                   | 25 m footprints along 8 tracks<br>non-repeating   | Canopy height (rh98), vertical profile, above-ground biomass.                                  | <b>Tried and abandoned.</b> Not a resolution problem — the footprint is far smaller than the fire. The problem is coverage: $\approx 1.4\%$ per pass, footprints never repeat, a thin pre-fire archive, and $n = 5$ after splitting by severity. Terrain adds $25\text{ m} \times \tan 23^\circ \approx 10.6\text{ m}$ to rh98. |
| ECOSTRESS            | Thermal on the ISS                             | 70 m, irregular                                   | Evapotranspiration, surface temperature.                                                       | Not used. Preprocessing orbit gives no reliable revisit.                                                                                                                                                                                                                                                                        |
| PRISMA / EnMAP       | Imaging spectrometers                          | 30 m, tasked                                      | Contiguous VNIR–SWIR spectra, including the 2.3 $\mu\text{m}$ CH <sub>4</sub> window.          | Not used. Tasked acquisitions; would have to be requested before the fire, which is the whole difficulty.                                                                                                                                                                                                                       |
| GHGSat               | Commercial CH <sub>4</sub> imager              | 25–50 m, tasked                                   | Point-source methane plumes.                                                                   | Not used. Commercial and tasked. This is the class of sensor my detectability calculation says would be needed.                                                                                                                                                                                                                 |
| GOSAT / GOSAT-2      | FTS sounder                                    | 10 km spots, sparse                               | Column CO <sub>2</sub> and CH <sub>4</sub> .                                                   | Not used. Sparse sampling, no imaging.                                                                                                                                                                                                                                                                                          |
| MethaneSAT           | Targeted CH <sub>4</sub> imager                | $\approx 100\text{ m} \times 400\text{ m}$        | Regional methane fluxes.                                                                       | Not used. Designed for oil and gas basins, not single fires.                                                                                                                                                                                                                                                                    |

#### 5 ANCILLARY DATASETS — NOT SATELLITES, BUT THEY CARRY THE ANSWER

| DATASET                | PROVIDER & RESOLUTION       | WHAT IT CONTRIBUTES                                                          | ROLE IN THE BUDGET                                                                          |
|------------------------|-----------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| CORINE Land Cover 2018 | EEA, 100 m, 44 classes      | Fuel <b>type</b> : broadleaf, coniferous, mixed, transitional woodland.      | Selects which land can burn and which fuel load D <sup>i</sup> applies.                     |
| ESA WorldCover 2020    | ESA, 10 m, 11 classes       | Fuel <b>extent</b> : where tree cover actually is, at Sentinel-2 resolution. | Intersected with CORINE. Removed 180.9 ha (28.3 %) of non-fuel inside the mapped perimeter. |
| Copernicus EMS EMSR449 | JRC rapid mapping, vector   | The operational burn perimeter, delineated by trained analysts.              | The independent perimeter truth against which my own severity mapping is checked.           |
| ESA CCI Biomass v6     | 100 m, above-ground biomass | Standing biomass density in t/ha.                                            | Cross-check on D = 118.9 t/ha standing.                                                     |
| ERA5-Land              | C3S reanalysis, 9 km hourly | Wind speed, temperature, humidity.                                           | Wind speed U in the plume residence time $\tau = L/U$ .                                     |

#### 6 BAND $\rightarrow$ INDEX $\rightarrow$ MEANING

| INDEX | FORMULA                                | WHY THOSE BANDS                                                                                                                               | WHAT A CHANGE IN IT MEANS                                                                    |
|-------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| NBR   | $(B8 - B12)/(B8 + B12)$                | B8 (842 nm) is high for healthy leaves; B12 (2190 nm) rises when water is removed and char is exposed. The contrast is maximal across a fire. | NBR falls after a fire; it recovers as the canopy returns.                                   |
| dNBR  | $NBR_{\text{pre}} - NBR_{\text{post}}$ | Differencing cancels anything constant between the two dates.                                                                                 | Magnitude of the change. Thresholds after Key & Benson (2006).                               |
| RBR   | $dNBR / (NBR_{\text{pre}} + 1.001)$    | Divides out how green the stand was beforehand.                                                                                               | Two stands that burned alike score alike (Parks 2014).                                       |
| NDMI  | $(B8A - B11)/(B8A + B11)$              | B11 (1610 nm) is a water-absorption band; B8A is the reference.                                                                               | Canopy moisture — used here to test whether the forest was already stressed before the fire. |
| NDVI  | $(B8 - B4)/(B8 + B4)$                  | Red is absorbed by chlorophyll, NIR is scattered by leaf structure.                                                                           | Greenness. Saturates in dense canopy, which is why NBR is preferred for severity.            |

**The honest sentence about GEDI.** “GEDI did not fail because its footprint is too coarse for a 391-hectare forest — a 25 m footprint is about 6 300 times smaller than the fire. It failed because it samples along eight narrow tracks that never repeat, so after splitting by severity class I was left with five usable shots, and on 23° slopes the terrain inside one footprint adds about 10 metres of apparent canopy height.”

## 7 PROCESSING LEVELS — THE GENERIC LADDER

| LEVEL | DEFINITION                     | WHAT HAS BEEN DONE TO THE DATA                                                                                                                     | CAN YOU DO SCIENCE WITH IT?            |
|-------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| L0    | Raw instrument packets         | Nothing. Telemetry as downlinked, with ancillary data attached, duplicates removed.                                                                | No.                                    |
| L1A   | Reconstructed, time-referenced | Unpacked and time-tagged. Radiometric and geometric coefficients are <i>appended but not applied</i> .                                             | No — still instrument counts.          |
| L1B   | Calibrated, geolocated         | Coefficients <i>applied</i> : counts → physical units (radiance, backscatter). Each sample has a lat/lon, but the grid is the instrument's own.    | Yes, with care.                        |
| L1C   | Resampled to a map grid        | Orthorectified and resampled onto a cartographic projection. For Sentinel-2 this is top-of-atmosphere reflectance in UTM tiles.                    | Yes — but still TOA.                   |
| L2    | Geophysical variable           | The atmosphere has been dealt with, or a physical retrieval has been run: surface reflectance, XCH <sub>4</sub> , FRP, LST. Same resolution as L1. | <b>Yes. This is the working level.</b> |
| L3    | Gridded / composited           | Mapped onto a uniform space–time grid; gaps filled by compositing over days or weeks.                                                              | Yes, for time series.                  |
| L4    | Modelled or assimilated        | Variables that were never measured directly — model output, data assimilation, inversions.                                                         | Yes, with the model's assumptions.     |

## 8 THE SAME LADDER, MISSION BY MISSION

| MISSION     | L0          | L1                                                                                                                         | L2 AND ABOVE                                                                                   | THE ONE I USED                                                                         |
|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Sentinel-1  | raw packets | SLC — complex, <b>phase preserved</b> , slant range (InSAR)<br>GRD — detected, multi-looked, ground range (amplitude only) | L2 OCN: ocean wind, wave, current                                                              | GRD, for the structural check that failed                                              |
| Sentinel-2  | raw packets | L1B radiance (instrument grid)<br>L1C TOA reflectance, orthorectified, 110 km UTM tiles                                    | L2A bottom-of-atmosphere reflectance + AOT + water vapour + SCL scene classification (Sen2Cor) | L2A — COPERNICUS/S2_SR_HARMONIZED                                                      |
| Sentinel-3  | raw packets | L1B calibrated radiances / brightness temperatures                                                                         | L2 LST, FRP, ocean colour, altimetry                                                           | none                                                                                   |
| Sentinel-5P | raw packets | L1B spectra: Earth radiance and solar irradiance                                                                           | L2 per species: L2__CH4__, CO, NO <sub>2</sub> , O <sub>3</sub> ...                            | L2 CH <sub>4</sub> — note Earth Engine <i>relabels</i> it L3_CH4 because it regrids it |
| MODIS       | raw packets | L1B calibrated radiances (MOD021KM)                                                                                        | L2 MOD14 fire mask + FRP<br>L3 MOD14A1 / MYD14A1 daily 1 km<br>L3 MCD64A1 monthly burned area  | MOD14A1 + MYD14A1 daily MaxFRP                                                         |

## 9 EVERY PRODUCT THIS PROJECT ACTUALLY OPENED

| PRODUCT                              | LEVEL                 | RESOLUTION | WHY THIS ONE                                           | WHAT IT PRODUCED                                                                      |
|--------------------------------------|-----------------------|------------|--------------------------------------------------------|---------------------------------------------------------------------------------------|
| COPERNICUS/S2_SR_HARMONIZED          | L2A                   | 10 / 20 m  | Surface reflectance, so NBR is comparable across dates | NBR, dNBR, RBR, NDMI, burnt area A <sup>i</sup>                                       |
| COPERNICUS/S2_SR_HARMONIZED SCL band | L2A                   | 20 m       | Per-pixel scene class from Sen2Cor                     | Cloud mask: classes 3, 8, 9, 10, 11 removed                                           |
| MODIS/061/MOD14A1, MYD14A1           | L3 daily              | 1 km       | Only free daily FRP with four looks per day            | FRE upper bound 102,976 t → duty cycle 14.5 %                                         |
| FIRMS (MODIS C6)                     | L3 daily              | 1 km       | Active-fire detection timing                           | Day-of-burning, fire progression                                                      |
| COPERNICUS/S5P/OFFL/L3_CH4           | L2 (regridded)        | 5.5 × 7 km | The only free daily methane column                     | Detectability test — 0 valid CH <sub>4</sub> days over the site; forward model 0.08 σ |
| COPERNICUS/CORINE/V20/100m/2018      | vector-derived raster | 100 m      | Fuel type classes                                      | β and D per class                                                                     |
| ESA/WorldCover/v100/2020             | L4 classification     | 10 m       | Fuel extent at Sentinel-2 scale                        | Removed 180.9 ha of non-fuel                                                          |
| Copernicus EMS EMSR449               | vector                | —          | Operational perimeter truth                            | 638.9 ha mapped perimeter                                                             |

## 9B THREE TRAPS HIDDEN IN THE WORD “LEVEL”

- L2 does not mean “ready”.** Sentinel-2 L2A is surface reflectance, but the Scene Classification Layer is a *guess* at what each pixel is. I still masked classes 3, 8, 9, 10 and 11 by hand and composited by median, because a single Sen2Cor misclassification over a burn scar propagates straight into dNBR.
- The same name can mean different levels in different catalogues.** Earth Engine serves the Sentinel-5P methane product as L3\_CH4 because it has been regridded, while ESA ships it as an L2 orbit product. The physics is unchanged; the geometry is not. If asked, say L2 retrieval, regridded.
- L3 daily is not one measurement.** MOD14A1 stores the *maximum* FRP observed in that 24 hours, from up to four overpasses. Integrating that maximum as if it were sustained all day is exactly the upper bound this project reports explicitly — and turns into the 14.5 % diurnal duty cycle.

## 10 ESA SNAP — WHAT IT IS

SNAP (Sentinel Application Platform) is ESA's free desktop toolbox, built from mission-specific toolboxes (S1TBX for radar, S2TBX and S3TBX for optical). Every operation is an **operator** in the Graph Processing Framework; operators chain into an XML graph that gpt runs from the command line, so a chain built once by hand becomes a batch job. The order below is not a preference — each step assumes the previous one has been done.

## 11 THE SENTINEL-1 PRE-PROCESSING CHAIN, IN THE ORDER IT MUST BE RUN

| #  | SNAP OPERATOR                                        | WHAT IT DOES PHYSICALLY                                                                                                                                                                                                                                                  | WHAT BREAKS IF YOU SKIP IT                                                                                                            |
|----|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>Apply-Orbit-File</b>                              | Replaces the predicted orbit state vectors in the product metadata with precise ones (POEORB, available ~20 days later).                                                                                                                                                 | Geolocation errors of metres to tens of metres; interferograms carry an orbital ramp.                                                 |
| 2  | <b>Remove-GRD-Border-Noise</b>                       | Deletes the invalid low-intensity samples the GRD processor leaves at the swath edges.                                                                                                                                                                                   | Dark artificial stripes at the image border bias any statistic you compute.                                                           |
| 3  | <b>ThermalNoiseRemoval</b>                           | Subtracts the additive instrument noise profile (NESZ) that the calibration annotation supplies, across range.                                                                                                                                                           | Low-backscatter targets — water, cross-polarised returns, burnt ground — read too bright.                                             |
| 4  | <b>Calibration</b><br>(radiometric)                  | Converts digital numbers to a physically meaningful radar quantity using the calibration LUT and the range-dependent gain: $\beta^0$ radar brightness, $\sigma^0$ backscatter per unit <i>ground</i> area, $\gamma^0$ per unit area perpendicular to the look direction. | Images from different dates, tracks or satellites are not comparable. Nothing quantitative is possible.                               |
| 5  | <b>Multilooking</b>                                  | Averages N independent looks in range and azimuth. Speckle is multiplicative and roughly $\sqrt{N}$ is gained in radiometric accuracy, paid for in resolution; also squares the pixels.                                                                                  | Speckle dominates; pixels stay rectangular, which distorts any later geometry.                                                        |
| 6  | <b>Speckle-Filter</b>                                | Adaptive spatial filtering — Lee, Refined Lee, Lee Sigma, Frost, Gamma-MAP, IDAN — that suppresses speckle while trying to preserve edges.                                                                                                                               | Class boundaries and change detection are drowned in coherent noise.                                                                  |
| 7  | <b>Terrain-Flattening</b>                            | <b>Radiometric</b> terrain correction. Uses a DEM to compute the true illuminated area of each resolution cell and converts $\beta^0$ to $\gamma^0_T$ .                                                                                                                  | On slopes the brightness is topography, not surface property. <b>This is the step whose absence broke my own Sentinel-1 analysis.</b> |
| 8  | <b>Terrain-Correction</b><br>(Range-Doppler)         | <b>Geometric</b> correction. Solves the range and Doppler equations together with the orbit and a DEM to place every SAR sample at its true ground position, then resamples to a map projection.                                                                         | Foreshortening, layover and shadow remain; the image cannot be overlaid on anything else.                                             |
| 9  | <b>Ellipsoid-Correction</b><br>(GG / average height) | Cheap geocoding against the ellipsoid at an assumed constant height, for flat terrain or when no DEM is available.                                                                                                                                                       | Acceptable on plains, unacceptable in the Apennines.                                                                                  |
| 10 | <b>LinearToFromdB</b>                                | $10 \cdot \log_{10}(\sigma^0)$ . Purely a display and statistics convenience.                                                                                                                                                                                            | Nothing physical, but histograms and differences behave badly in linear power.                                                        |
| 11 | <b>Subset · Collocate · BandMaths · Reproject</b>    | Housekeeping: clip to the AOI, put two products on one grid, compute an index, change CRS.                                                                                                                                                                               | —                                                                                                                                     |

## 11B THE SAME CHAIN AS A GPT COMMAND

```
gpt Apply-Orbit-File -Ssource=S1A_IW_GRDH_...zip -t 01_orb.dim
gpt Remove-GRD-Border-Noise -Ssource=01_orb.dim -t 02_bn.dim
gpt ThermalNoiseRemoval -Ssource=02_bn.dim -t 03_tnr.dim
gpt Calibration -PoutputBetaBand=true -PoutputSigmaBand=false -Ssource=03_tnr.dim -t 04_cal.dim
gpt Terrain-Flattening -PdemName='SRTM 1Sec HGT' -Ssource=04_cal.dim -t 05_tf.dim
gpt Terrain-Correction -PmapProjection=EPSG:3035 -PpixelSpacingInMeter=20 -Ssource=05_tf.dim -t 06_tc.dim
```

Note the `-PoutputBetaBand=true`: terrain flattening consumes  $\beta^0$ , not  $\sigma^0$ . Getting that flag wrong is the commonest way to produce a chain that runs cleanly and returns physically meaningless numbers.

## 11C WHICH CHAIN ANSWERS WHICH QUESTION

| THE QUESTION                                                 | PRODUCT TO ORDER                     | CHAIN                                                                                                                    |
|--------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| How bright is the surface to radar, comparably across dates? | GRD                                  | orbit → border → noise → <b>calibration</b> → speckle → terrain flattening → terrain correction                          |
| Did the ground move?                                         | SLC, same track, same pass direction | split → orbit → coregistration → ESD → interferogram → deburst → topo removal → filter → unwrap → displacement → geocode |
| Did the structure change — e.g. did a canopy burn away?      | SLC pair before and after            | as above, but stop at <b>coherence</b> : no unwrapping needed, and it is immune to radiometric errors                    |
| What burned, and how severely?                               | Sentinel-2 L2A                       | no SNAP required — cloud mask, composite, band maths for NBR/dNBR/RBR                                                    |

## 12 THE INTERFEROMETRIC (INSAR) CHAIN

| #  | SNAP OPERATOR                                     | WHAT IT DOES                                                                                                                        |
|----|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>TOPSAR-Split</b>                               | Selects one sub-swath and the burst range that covers the AOI. IW products are three sub-swaths of nine bursts each.                |
| 2  | <b>Apply-Orbit-File</b>                           | Precise orbits, as above — mandatory before coregistration.                                                                         |
| 3  | <b>Back-Geocoding</b>                             | DEM-assisted coregistration: maps every slave pixel onto the master grid to a small fraction of a pixel.                            |
| 4  | <b>Enhanced-Spectral-Diversity</b>                | Refines the azimuth shift for TOPS acquisitions. Without it the phase jumps at every burst boundary.                                |
| 5  | <b>Interferogram</b>                              | Master × conjugate(slave). Outputs the interferometric phase and the <b>coherence</b> , which is the correlation magnitude.         |
| 6  | <b>TOPSAR-Deburst</b>                             | Merges the bursts into one continuous image, removing the demarcation lines.                                                        |
| 7  | <b>TopoPhaseRemoval</b>                           | Subtracts the flat-Earth phase and the topographic phase predicted from a DEM, leaving deformation plus noise.                      |
| 8  | <b>GoldsteinPhaseFiltering</b>                    | Adaptive power-spectrum filtering that sharpens the fringes so unwrapping can succeed.                                              |
| 9  | <b>SNAPHU export → unwrap → import</b>            | Resolves the $2\pi$ ambiguity: turns wrapped phase in $[-\pi, \pi]$ into a continuous field.                                        |
| 10 | <b>Phase-to-Displacement / Phase-to-Elevation</b> | Converts unwrapped phase to line-of-sight displacement ( $\lambda/4\pi$ per radian) or to height, using the perpendicular baseline. |
| 11 | <b>Range-Doppler Terrain-Correction</b>           | Final geocoding of the result.                                                                                                      |

The three InSAR limits to name out loud: temporal decorrelation (vegetation moves between passes, so coherence collapses — which is exactly why coherence *loss* is itself a fire signal); the critical baseline, beyond which the two geometries no longer share spectral content; and the atmospheric phase screen, where water vapour masquerades as centimetres of ground motion.

## 13 TWO OPERATORS WORTH BEING ABLE TO EXPLAIN ON A BLACKBOARD

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>RADIOMETRIC CALIBRATION — WHAT IT ACTUALLY REMOVES</b></p> <p>The detector returns a digital number. That number contains the antenna pattern, the range spreading loss (power falls as <math>R^4</math> for a distributed target), the processor scaling and the receiver gain — none of which are properties of the ground. Calibration divides all of that out using the calibration LUT shipped with the product.</p> <p><math>\sigma^0 =  DN ^2 / A\sigma^2</math> — A from the annotation LUT</p> <p>Afterwards two images taken from different ranges, different tracks or different satellites can be compared. The choice among <math>\beta^0</math>, <math>\sigma^0</math> and <math>\gamma^0</math> is a choice of reference area: the slant plane, the ground, or the plane perpendicular to the look direction.</p> | <p><b>RANGE-DOPPLER TERRAIN CORRECTION — THE GEOMETRY</b></p> <p>A SAR image is not a picture. Each sample is addressed by <b>slant range</b> and <b>azimuth time</b>, not by latitude and longitude. To place it on the ground you solve two equations simultaneously for each DEM cell:</p> $\begin{cases} R =  P_{\text{sat}}(t) - P_{\text{gnd}}  & (\text{range}) \\ f_D = -2(v_{\text{sat}} - v_{\text{gnd}}) \cdot (P_{\text{sat}} - P_{\text{gnd}}) / (\lambda R) & (\text{Doppler}) \end{cases}$ <p>The DEM supplies the ground height, the orbit supplies <math>P_{\text{sat}}(t)</math>. Solving them corrects <b>foreshortening</b> (slopes facing the radar compress), <b>layover</b> (a peak returns before its base) and <b>shadow</b> (steep back-slopes return nothing at all).</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14 THE SENTINEL-1 LESSON FROM THIS PROJECT

I ran a Sentinel-1 backscatter check on the burnt area and obtained **+1.97 dB after the fire** — physically the wrong sign, since removing a canopy should lower volume scattering. Two causes, both of them SNAP steps:

1. **No terrain flattening.** On the 23° slopes around Arischia,  $\sigma^0$  is modulated by local incidence angle far more strongly than by the fire. Step 7 of the chain on the previous page is not optional in the Apennines.
2. **Ascending and descending tracks were mixed.** They see opposite slope faces, so their backscatter distributions are not the same population. Separating the tracks alone removed 77 % of the anomaly, leaving +0.46 dB.

The honest conclusion I report: amplitude was the wrong observable. **Interferometric coherence** is a direct structural-change measurement that does not depend on radiometric calibration at all, and it is the recommended replacement — reprocessed from L1 SLC, with precise orbits, thermal noise removal, terrain flattening to  $\gamma^0_T$ , and single-track stacking.

## 15 OPTICAL OPERATORS IN SNAP

| OPERATOR                  | FUNCTION                                                                                                                               |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Resample</b>           | Puts the 10, 20 and 60 m Sentinel-2 bands on one grid. Nothing else in SNAP will run on a multi-resolution product until this is done. |
| <b>Sen2Cor</b>            | L1C → L2A: atmospheric correction, terrain and cirrus correction, and the Scene Classification Layer used here for cloud masking.      |
| <b>Idepix</b>             | Alternative pixel classification — cloud, shadow, snow — when SCL is not trusted.                                                      |
| <b>BandMaths</b>          | Arbitrary per-pixel expressions. This is where NBR, dNBR and RBR are written.                                                          |
| <b>BiophysicalOp</b>      | Neural-network retrieval of LAI, FAPAR, FCOVER from Sentinel-2 reflectance.                                                            |
| <b>Collocate</b>          | Puts two products of different geometry on a common grid so they can be differenced.                                                   |
| <b>Reproject / Subset</b> | CRS change and spatial clipping. Areas belong in an equal-area CRS — EPSG:3035 here, never EPSG:4326.                                  |